Scorso for Malnutrition Risk Screening
Gelderse Vallei

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1. Synopsis – Scorso for Malnutrition Risk Screening - Gelderse Vallei

IMPORTANT	**THIS IS A PRODUCT TEST PROTOCOL** **THIS IS NOT A CINICICAL INVESTIGATION**
Title	**Scor**so **E**valuation for malnutrition risk screening
Acronym	Proposal: SCORE
Partner Site	Prof. Dr. ARH (Arthur) van Zanten Intensivist and medical head of ICU Gelderse Vallei Hospital
Department	Intensive Care Unit and / or a general ward
Duration of the Test	The technical integration phase will start in xxxxx 2024. The maximum duration of the usability phase of the test will be 2 months of SCORSO use by the clinical team.
Aim	To evaluate the usability of mobile solutions for the screening of malnutrition risk of hospitalized patients, as well as the integration & implementation process of such monitoring tools within an hospital setting.
Introduction & Rationale	Malnutrition is a common finding in hospitalized patients, concerning more than 30% of patients entering the hospital. Malnutrition is associated with increased risk of complications, length of stay, mortality, and higher treatment costs. Existing recommendations and guidelines for nutrition support are often not sufficiently or inadequately applied. The under-enforced application of guidelines may be partly explained by the physicians' lack of specific training in artificial nutrition and/or by the manual work and calculation. The use of computer-based decision-support systems has been shown to improve the quality of nutrition orders. [1,2,3,4,5,6,7,8] Nutrition screening is the first and essential step – as described in the ASPEN Nutrition Care Pathways - to identify individuals at risk for Nutrition in Clinical Practices. The scoring result triggers the procedure of treatment including assessment, nutrition targets setting, prescription, and follow-up. SCORSO® is a mobile application to screen the malnutrition risk of patients. It helps Health Care Professionals (HCP) detect malnutrition risk, in a simpler and more reliable way. Scorso can be used standalone or in combination with NUTROW® NUTROW® is a clinical nutrition management system that integrates and organizes all nutritionally relevant patient data coming from multiple sources such as electronic health records (EHR), laboratories and feeding pumps in one centralized software.

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	The purpose of the test is to evaluate the implementation and usability of SCORSO mobile application in an ICU and/or a general ward and collect data on improvement opportunities. SCORSO is for professional use only. 1 Jennifer C. North et al, Nutrition Informatics Applications in Clinical Practice: a Systematic Review. AMIA Annu Symp Proc. 2015; 2015: 963–972 2 Florence Ettori et al. Impact of a computer-assisted decision support system (CDSS) on nutrition management in critically ill hematology patients: the NUTCHOCO study (nutritional care in hematology oncologic patients and critical outcome). Ann. Intensive Care (2019) 9:53 3 North JC, Jordan KC, Metos J, Hurdle JF. Nutrition Informatics Applications in Clinical Practice: a Systematic Review. AMIA Annu Symp Proc. 2015 Nov 5;2015:963-72 4 van Schijndel RJ et al. Computer-aided support improves early and adequate delivery of nutrients in the ICU. Neth J Med. 2009 Dec;67(11):388-93. 5 Conseil M et al. A simple widespread computer helps improve nutrition support orders and decreases infection complications in critically ill patients. PLoS One. 2013 May 30;8(5) 6 MM Berger at al. Impact of a computerized information system on quality of nutritional support in the ICU. Nutrition. 2006 Mar;22(3):221-9. 7 Bally MR et al. Nutritional Support and Outcomes in Malnourished Medical Inpatients: A Systematic Review and Meta-analysis. JAMA Intern Med. 2016 Jan;176(1):43-53. 8 Llido LO. The impact of computerization of the nutrition support process on the nutrition support program in a tertiary care hospital in the Philippines: report for the years 2000-2003. Clin Nutr. 2006 Feb;25(1):91-101.
Benefit-Risk Assessment	This is not a clinical study; patients will not directly be impacted in their daily care and will not directly benefit from this test either. From this test, Gelderse Vallei can assess how SCORSO is able to improve patient care in terms of determining malnutrition risk in an easier, faster and more reliable way. In turn, Scorso can also be optimized to reach this goal. Having such knowledge, physicians will be able to better tailor the treatment as it is needed for the patient, respectively, resulting in the best available outcome, e.g. shorter length of stay at ICU or hospital in general, less infections, and/or increased muscle strength in comparison to standard procedures.
Objectives	**Objective 1** - Evaluate the technical integration and implementation of SCORSO within Gelderse Vallei IT **Objective 2** - Evaluate the implementation and use of SCORSO by healthcare professionals in the target ward **Objective 3** - Collect suggestions for improving Dim3 products

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Method	Each clinical user will be requested to use SCORSO to screen the nutrition risk of their patients according to their internal Standard Operating procedure. At the end of each phase, each user will be requested to fill in a user-specific satisfaction questionnaire.
Assessment criteria	Assessment criteria for objectives 1 & 3 - Evaluate – by a questionnaire - the satisfaction level of Gelderse Vallei IT technical team regarding the integration and implementation of SCORSO with hospital IT systems - The questionnaire – to be filled, at the end of this phase, by the professionals who have been working on the implementation - will include: ○ A satisfaction section (closed questions) including a qualitative analysis of time spent by the IT team ○ And will collect improvement suggestions (open questions) - Lower satisfaction limit set at 70% Note: The user rights management is out of scope of the integration. Assessment criteria for objectives 2 & 3 - Evaluate – by a questionnaire - the usability and satisfaction level of the clinical team (dietitians, nurses) regarding the training and use of SCORSO - The questionnaire – to be filled, throughout and at the end of the test, by the professional who have been using SCORSO - will include: ○ A satisfaction section (closed questions) including a qualitative analysis of time spent using SCORSO ○ And will collect improvement suggestions (open questions) - Lower satisfaction limit set at 70%
Persons involved in the Test	Objective 1 & 3 - Personnel in Gelderse Vallei IT department working in the integration & implementation of SCORSO with hospital IT systems Objectives 2 & 3 - Dietitians and nurses working in the target ward
Endpoints	Objective 1 At the end of the integration - Duration of the integration - Satisfaction score
	Objective 2 During the use of SCORSO (at least 2 months) and at the end of the test - Duration of the test - Number of persons involved - Satisfaction score
	Objective 3 - Improvements proposals based on the suggestions formulated in the questionnaires

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Proposed test steps	1. SCORSO installation and integration with IT systems of the hospital, except for users' rights management 2. Training of Dietitians and Nurses in the target ward 3. Questionnaire to IT personnel 4. Usage in the target ward – Maximum 2 months 5. Questionnaire to clinical personnel at different stages of the trial *Note: The target ward should be defined before step 1.* *The integration & implementation planning of SCORSO in target ward will be defined after discussion with clinical and technical staff of Gelderse Vallei hospital (step 1 to 3).* *Note: An intermediary report will be issued.*
Data Protection	Third parties will not receive patient-related data at any time. Data will only be analyzed in an anonymized manner. Results will be published only anonymized in a way that makes assignment or backtracking of individual data impossible.
Publication	The test results might be published in an appropriate scientific journal or used for product presentation.
Financing	

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2. Test Schedule

	Technical integration	Technical questionnaire End of integration	Usability Clinical questionnaire Month 1	Usability Clinical questionnaire End
Determination of target ward	X			
SCORSO installation & integration with IT systems	X	X		
Training of clinical personnel			X	
Screening tool			X	X

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3. Signatures

I have read and fully understand the requirements and conditions of this test and confirm hereby conducting this test according to the present description.

Moreover, I accept to the following terms:

- Any proposed change in the test will be submitted to the steering committee for approval.
- Keep test content confidential.
- Any formal term required for this type of test.

With my signature I hereby confirm receipt of the test description.

		Signature
Principal Investigator *Name* *Department*	**Prof. Dr. ARH (Arthur) van Zanten** **Intensivist and medical head of ICU - Gelderse Vallei Hospital**	
Project Manager *Name* *Department*		
Title *Name* *Department*		

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